

机器视觉技术

霍夫变换

Hough Transform — 中文翻译版（含原图）

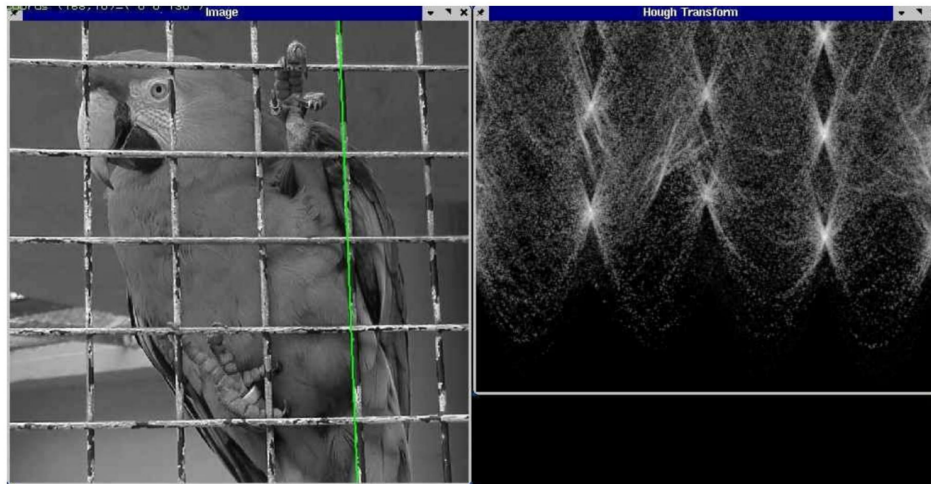
共 32 页 | 翻译时间：2026年6月

Fitting

(Hough transform)

拟合
(霍夫变换)

Fitting: The Hough transform



Source: S. Lazebnik

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拟合：霍夫变换

来源：S. Lazebnik

Voting schemes

- Let each feature vote for all the models that are compatible with it
- Hopefully the noise features will not vote consistently for any single model
- Missing data doesn't matter as long as there are enough features remaining to agree on a good model

Source: S. Lazebnik

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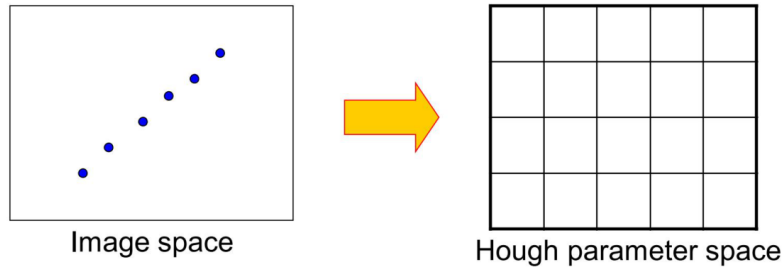
投票机制

- 让每个特征为所有与其兼容的模型投票
- 希望噪声特征不会对任何单一模型产生一致的投票
- 只要有足够多的特征剩余并一致认同一个良好模型，缺失数据不会造成影响

来源: S. Lazebnik

Hough transform

- An early type of voting scheme
- General outline:
 - Discretize parameter space into bins
 - For each feature point in the image, put a vote in every bin in the parameter space that could have generated this point
 - Find bins that have the most votes



P.V.C. Hough, *Machine Analysis of Bubble Chamber Pictures*, Proc. Int. Conf. High Energy Accelerators and Instrumentation, 1959

Source: S. Lazebnik

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霍夫变换

- 一种早期的投票机制
- 总体流程：
 - 将参数空间离散化为区间 (bin)
 - 对于图像中的每个特征点，在参数空间中可能生成该点的每个区间中投一票
 - 找出获得最多投票的区间

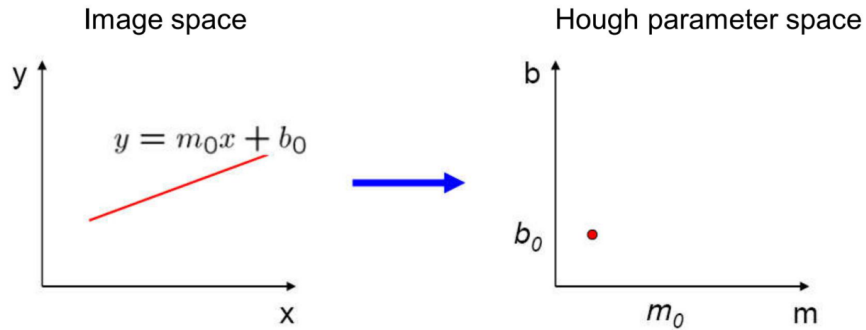
图像空间 霍夫参数空间

P.V.C. Hough, *Machine Analysis of Bubble Chamber Pictures*, Proc. Int. Conf. High Energy Accelerators and Instrumentation, 1959

来源: S. Lazebnik

Parameter space representation

- A line in the image corresponds to a point in Hough space



Source: S. Seitz

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参数空间表示

- 图像空间中的一条直线对应霍夫空间中的一个点

图像空间 霍夫参数空间

$$y = m_0x + b_0$$

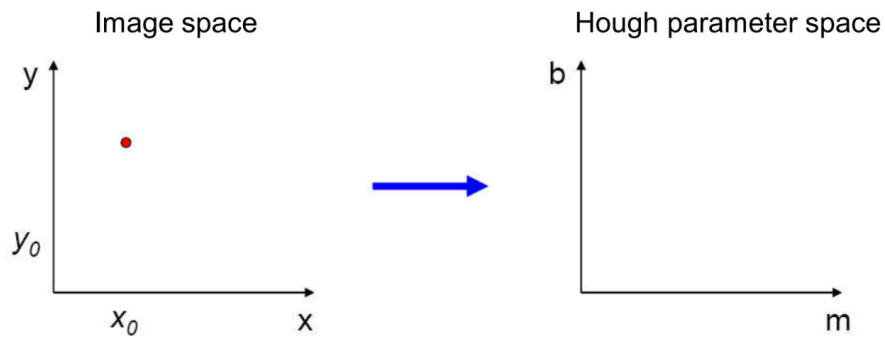
$y \rightarrow$ 图像空间 $b \rightarrow$ 参数空间

$m_0 \rightarrow m$

来源: S. Seitz

Parameter space representation

- What does a point (x_0, y_0) in the image space map to in the Hough space?



Source: S. Lazebnik

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参数空间表示

- 图像空间中的一个点 (x_0, y_0) 映射到霍夫空间中的什么?

图像空间 霍夫参数空间

$y \rightarrow$ 图像空间 $b \rightarrow$ 参数空间

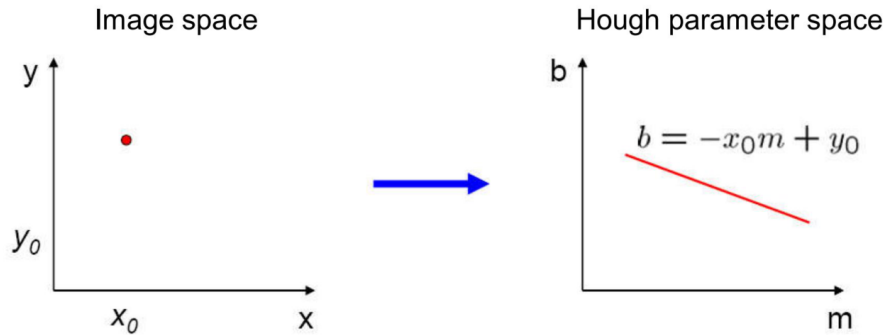
$y_0 \bullet$

$x_0 \rightarrow x \rightarrow m$

来源: S. Lazebnik

Parameter space representation

- What does a point (x_0, y_0) in the image space map to in the Hough space?
 - Answer: the solutions of $b = -x_0m + y_0$
 - This is a line in Hough space



Source: S. Lazebnik

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参数空间表示

- 图像空间中的一个点 (x_0, y_0) 映射到霍夫空间中的什么?
- 答案: 方程 $b = -x_0m + y_0$ 的解
- 这在霍夫空间中是一条直线

图像空间 霍夫参数空间

$y \rightarrow$ 图像空间 $b \rightarrow$ 参数空间

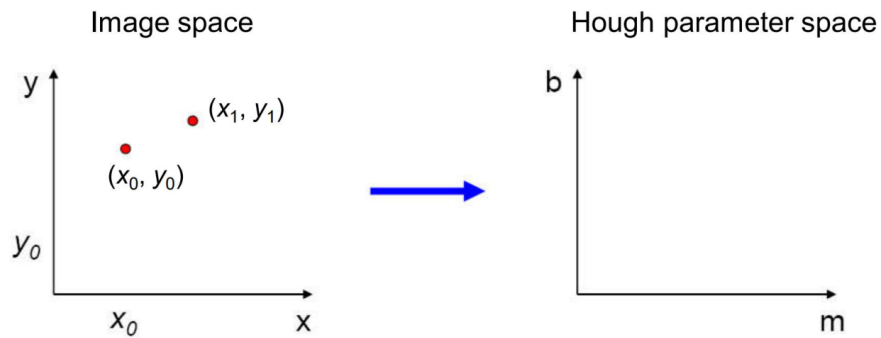
$y_0 \bullet b = -x_0m + y_0$

$x_0 \rightarrow x \rightarrow m$

来源: S. Lazebnik

Parameter space representation

- Where is the line that contains both (x_0, y_0) and (x_1, y_1) ?



Source: S. Lazebnik

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参数空间表示

- 同时包含点 (x_0, y_0) 和 (x_1, y_1) 的直线在哪里?

图像空间 霍夫参数空间

$y \rightarrow$ 图像空间 $b \rightarrow$ 参数空间

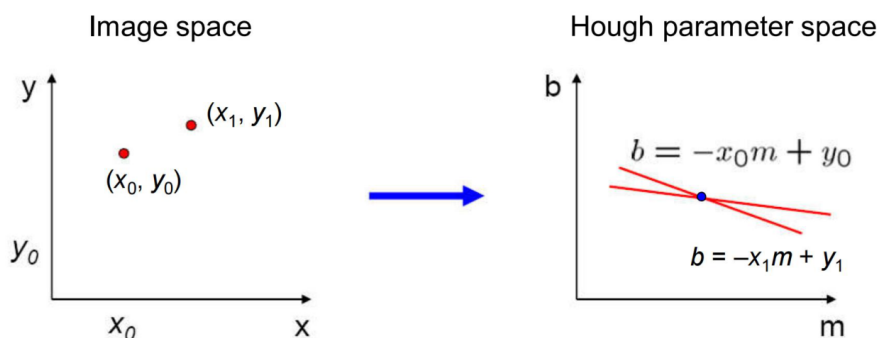
$\bullet (x_1, y_1)$

$(x_0, y_0) \bullet \rightarrow \rightarrow m$

来源: S. Lazebnik

Parameter space representation

- Where is the line that contains both (x_0, y_0) and (x_1, y_1) ?
 - It is the intersection of the lines $b = -x_0m + y_0$ and $b = -x_1m + y_1$



Source: S. Lazebnik

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参数空间表示

- 同时包含点 (x_0, y_0) 和 (x_1, y_1) 的直线在哪里?
- 答案是直线 $b = -x_0m + y_0$ 与 $b = -x_1m + y_1$ 的交点

图像空间 霍夫参数空间

$y \rightarrow$ 图像空间 $b \rightarrow$ 参数空间

$\bullet (x_1, y_1) \rightarrow b = -x_1m + y_1$

$(x_0, y_0) \bullet \rightarrow b = -x_0m + y_0$

$\rightarrow m$

来源: S. Lazebnik

Parameter space representation

- Problems with the (m,b) space:
 - Unbounded parameter domain
 - Vertical lines require infinite m

Source: S. Lazebnik

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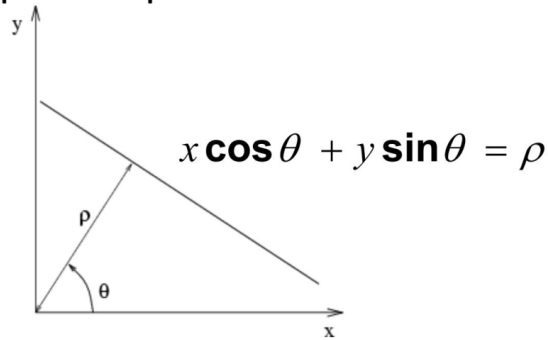
参数空间表示

- (m, b) 空间的问题:
 - 参数域无界
 - 垂直线需要无限的 m 值

来源: S. Lazebnik

Parameter space representation

- Problems with the (m,b) space:
 - Unbounded parameter domain
 - Vertical lines require infinite m
- Alternative: polar representation



Each point will add a sinusoid in the (θ, ρ) parameter space

Source: S. Lazebnik

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参数空间表示

- (m, b) 空间的问题:
 - 参数域无界
 - 垂直线需要无限的 m 值
- 替代方案: 极坐标表示

$$x \cos \theta + y \sin \theta = \rho$$

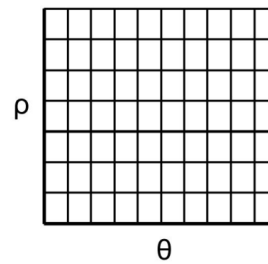
每个点在 (θ, ρ) 参数空间中添加一条正弦曲线

来源: S. Lazebnik

Algorithm outline

- Initialize accumulator H to all zeros
- For each edge point (x,y) in the image
 - For $\theta = 0$ to 180
 - $\rho = x \cos \theta + y \sin \theta$
 - $H(\theta, \rho) = H(\theta, \rho) + 1$
 - end
- end
- Find the value(s) of (θ, ρ) where $H(\theta, \rho)$ is a local maximum
 - The detected line in the image is given by
 - $\rho = x \cos \theta + y \sin \theta$

H: accumulator array (votes)



Source: S. Lazebnik

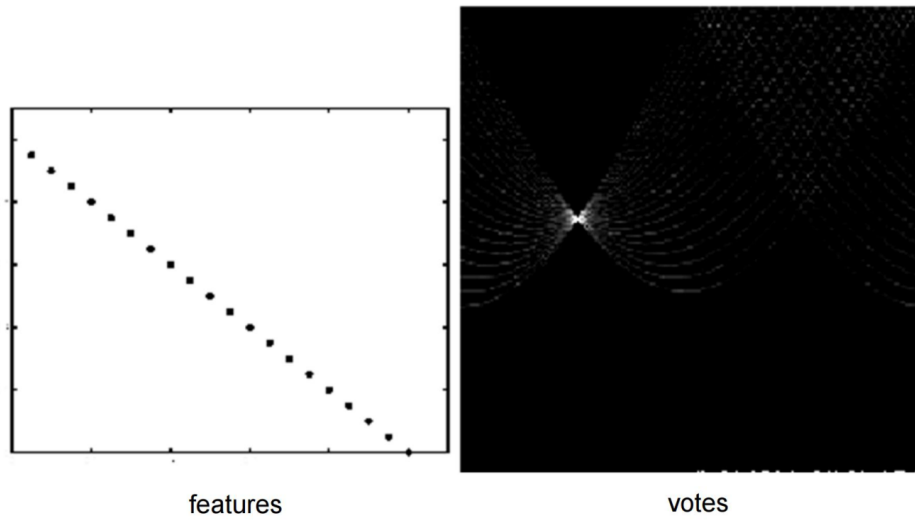
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算法流程

- 将累加器 H 初始化为全零 H: 累加器数组 (投票)
- 对于图像中的每个边缘点 (x,y)
 - 对于 $\theta = 0$ 到 180
 - $\rho = x \cos \theta + y \sin \theta$
 - $H(\theta, \rho) = H(\theta, \rho) + 1$
 - 结束
- 结束
- 找出 $H(\theta, \rho)$ 为局部最大值的 (θ, ρ) 值
- 图像中检测到的直线由下式给出:
 - $\rho = x \cos \theta + y \sin \theta$

来源: S. Lazebnik

Basic illustration



Source: S. Lazebnik

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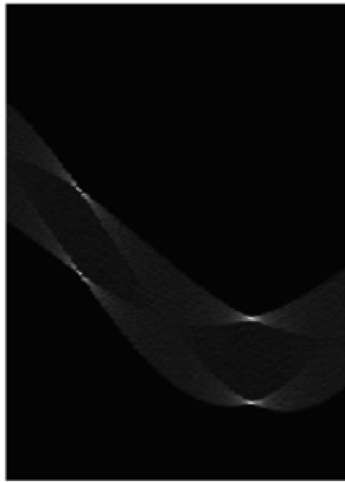
基本示例

该幻灯片展示霍夫变换的基本原理：左侧为图像空间中的特征点，右侧为参数空间中的投票结果。参数空间中亮度最高的点对应图像空间中检测到的直线。

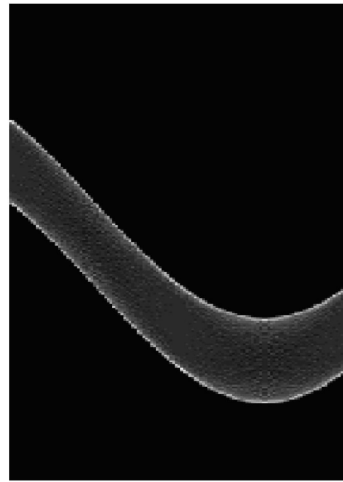
来源：S. Lazebnik

Other shapes

Square



Circle



Source: S. Lazebnik

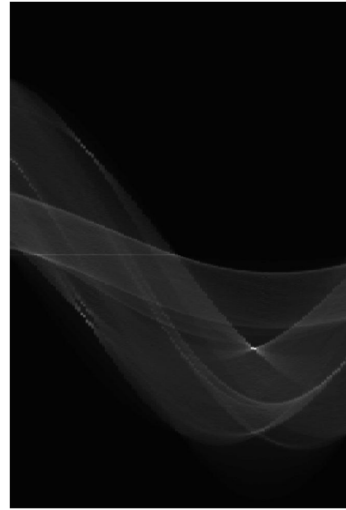
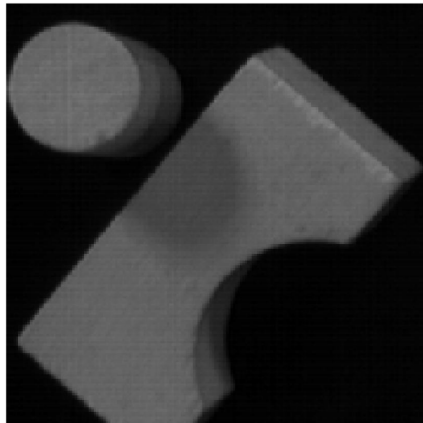
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其他形状

该幻灯片展示霍夫变换可以检测的其他形状，包括正方形和圆形。不同形状在参数空间中有不同的投票模式。

来源: S. Lazebnik

Several lines



Source: S. Lazebnik

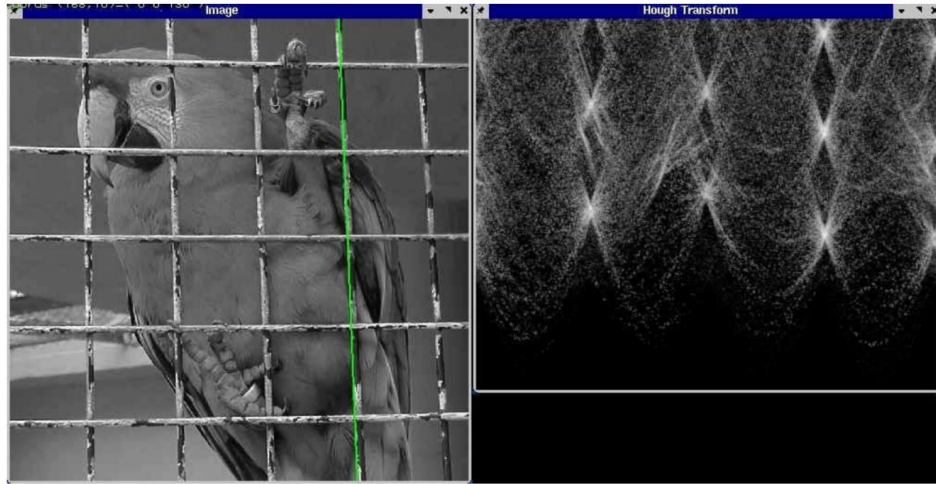
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多条直线

该幻灯片展示霍夫变换如何同时检测图像中的多条直线。参数空间中的多个局部峰值对应图像中的多条直线。

来源: S. Lazebnik

A more complicated image



http://ostatic.com/files/images/ss_hough.jpg

Source: S. Lazebnik

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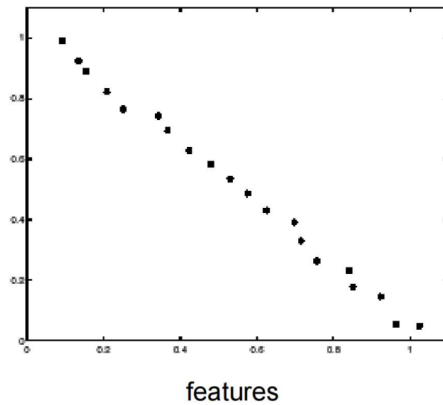
更复杂的图像

该幻灯片展示在一张更复杂的真实图像上应用霍夫变换的效果。左侧为原始图像，右侧为霍夫变换检测结果。

http://ostatic.com/files/images/ss_hough.jpg

来源: S. Lazebnik

Effect of noise



Source: S. Lazebnik

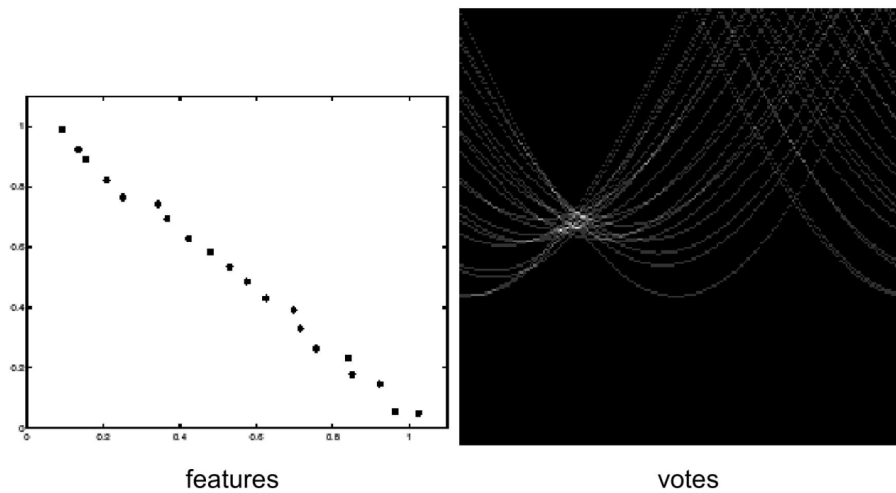
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噪声的影响

该幻灯片展示噪声对霍夫变换的影响：左侧为带有噪声的特征点，噪声使得参数空间中的投票变得分散。

来源：S. Lazebnik

Effect of noise



Peak gets fuzzy and hard to locate

Source: S. Lazebnik

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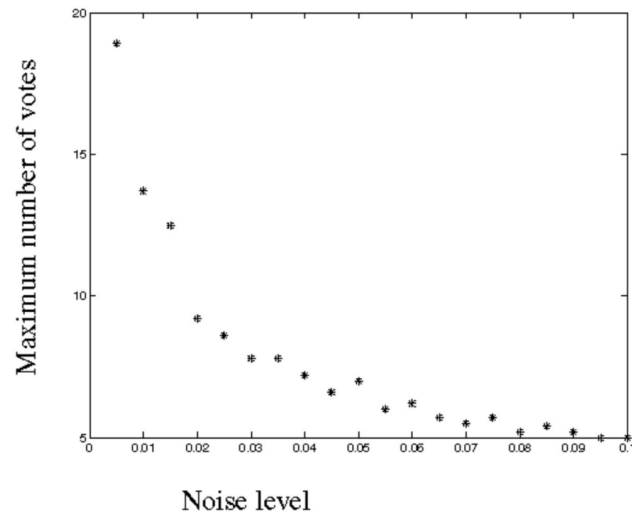
噪声的影响

- 左侧：特征点 右侧：投票结果
- 峰值变得模糊且难以定位

来源：S. Lazebnik

Effect of noise

- Number of votes for a line of 20 points with increasing noise:



Source: S. Lazebnik

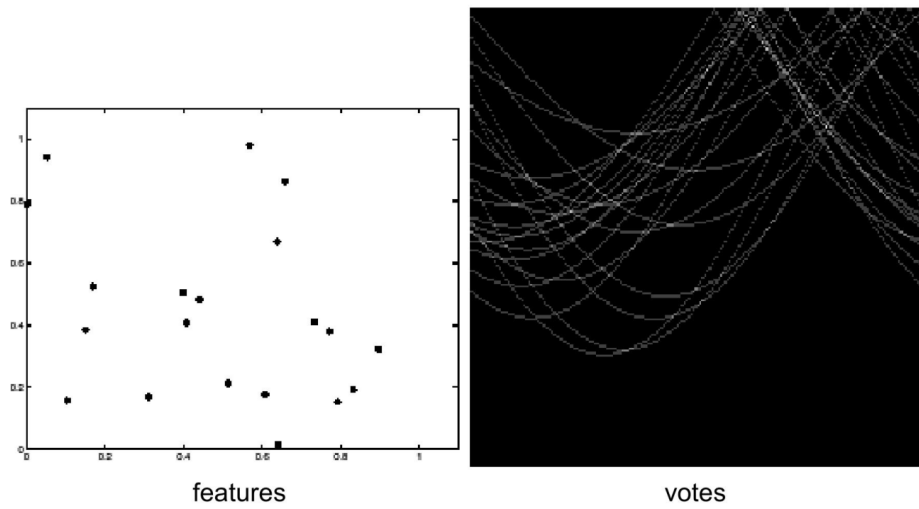
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噪声的影响

- 一条由20个点组成的直线在不同噪声水平下的投票数：
 - 随着噪声水平增加，最大投票数减少
 - 该幻灯片展示了投票数随噪声水平变化的图表

来源：S. Lazebnik

Random points



Uniform noise can lead to spurious peaks in the array

Source: S. Lazebnik

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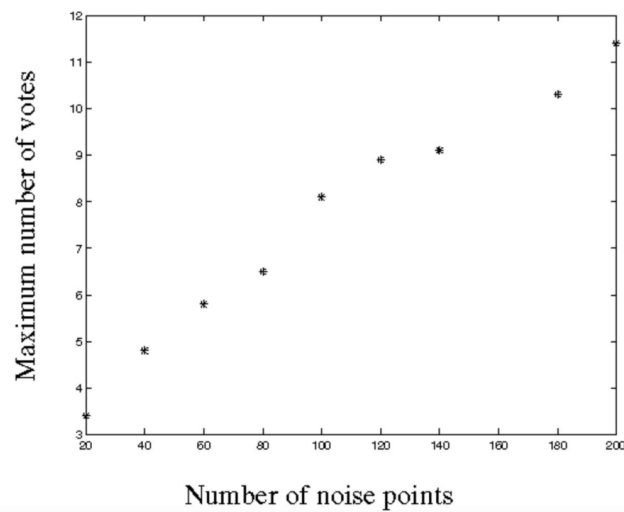
随机点

- 左侧：特征点（随机分布） 右侧：投票结果
- 均匀噪声可能导致累加器数组中出现虚假峰值

来源：S. Lazebnik

Random points

- As the level of uniform noise increases, the maximum number of votes increases too:



Source: S. Lazebnik

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随机点

- 随着均匀噪声水平的增加，最大投票数也在增加：
 - 该幻灯片展示了最大投票数随噪声点数量变化的图表
 - 随机噪声点可能偶然形成投票峰值

来源: S. Lazebnik

Dealing with noise

- Choose a good grid / discretization
 - Too coarse: large votes obtained when too many different lines correspond to a single bucket
 - Too fine: miss lines because some points that are not exactly collinear cast votes for different buckets
- Increment neighboring bins (smoothing in accumulator array)
- Try to get rid of irrelevant features
 - Take only edge points with significant gradient magnitude

Source: S. Lazebnik

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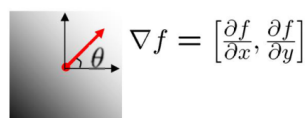
处理噪声

- 选择合适的网格/离散化粒度
 - 太粗糙：大量不同的直线对应同一个区间，获得大量投票
 - 太精细：某些不完全共线的点会投票给不同的区间，导致漏检直线
- 对相邻区间进行增量投票（累加器数组平滑）
- 尝试去除不相关的特征
 - 只取具有显著梯度幅值的边缘点

来源：S. Lazebnik

Incorporating image gradients

- Recall: when we detect an edge point, we also know its gradient direction
- But this means that the line is uniquely determined!
- Modified Hough transform:



$$\theta = \tan^{-1} \left(\frac{\partial f / \partial y}{\partial f / \partial x} \right)$$

```

For each edge point (x,y)
    θ = gradient orientation at (x,y)
    ρ = x cos θ + y sin θ
    H(θ, ρ) = H(θ, ρ) + 1
end
  
```

Source: S. Lazebnik

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结合图像梯度

- 回顾：当我们检测到一个边缘点时，我们也知道其梯度方向：

$$\nabla f = [\partial f / \partial x, \partial f / \partial y]$$

$$\theta = \tan^{-1} (\partial f / \partial y / \partial f / \partial x)$$

- 但这意味着直线被唯一确定！

- 改进的霍夫变换：

对于每个边缘点 (x,y)

θ = 该点 (x,y) 处的梯度方向

$$\rho = x \cos \theta + y \sin \theta$$

$$H(\theta, \rho) = H(\theta, \rho) + 1$$

结束

来源: S. Lazebnik

Hough transform for circles

- How many dimensions will the parameter space have?
- Given an oriented edge point, what are all possible bins that it can vote for?

Source: S. Lazebnik

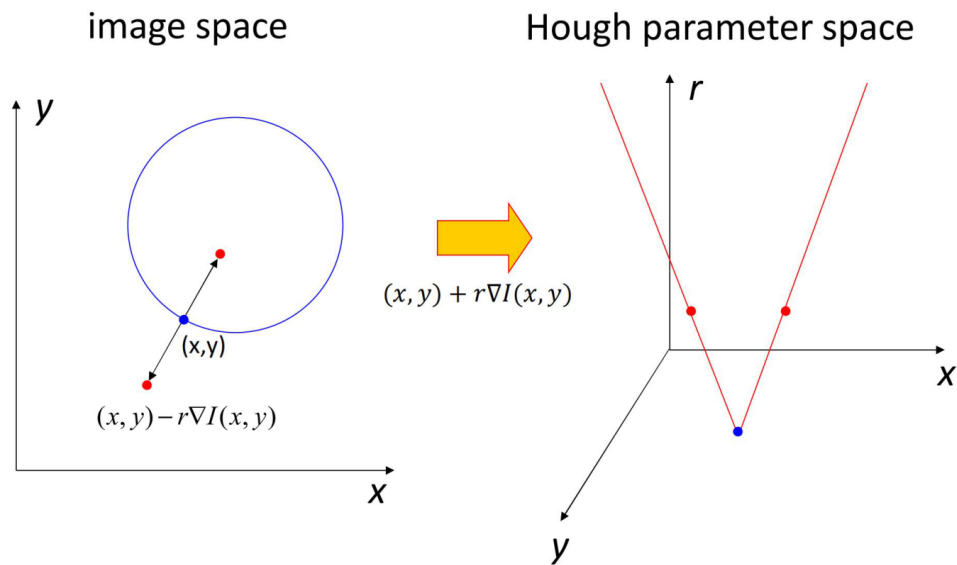
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圆的霍夫变换

- 参数空间的维度是多少?
- 给定一个有方向的边缘点，它可以投票给哪些可能的区间?

来源: S. Lazebnik

Hough transform for circles



Source: S. Lazebnik

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圆的霍夫变换

该幻灯片展示圆的霍夫变换原理图：

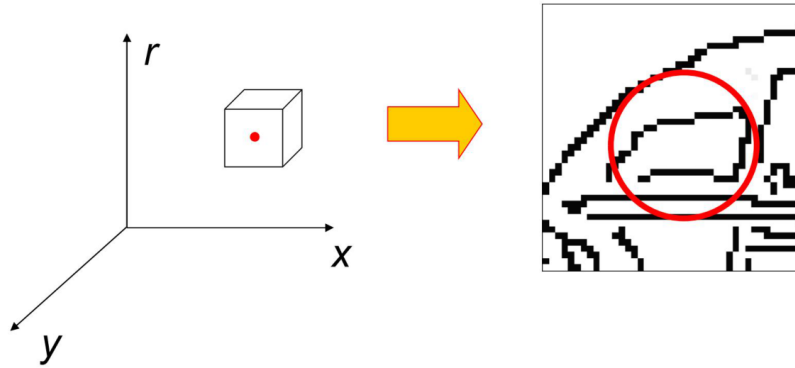
图像空间 霍夫参数空间

- 图像空间中的圆 (x, y) 映射到参数空间
- 已知梯度方向时, $(x, y) + r \cdot \nabla I(x, y)$ 指向圆心

来源: S. Lazebnik

Hough transform for circles

- Conceptually equivalent procedure: for each (x, y, r) , draw the corresponding circle in the image and compute its “support”



Is this more or less efficient than voting with features?

Source: S. Lazebnik

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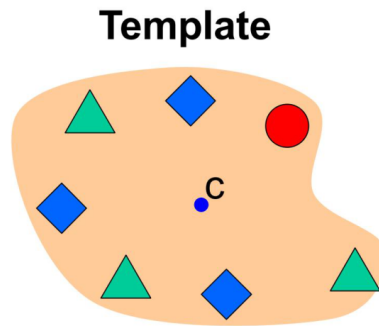
圆的霍夫变换

- 概念上等价的过程：对于每个 (x, y, r) ，在图像中画出相应的圆并计算其“支持度”
- 该幻灯片展示了这种方法的示意图
- 这种方法比用特征投票更高效还是更低效？

来源：S. Lazebnik

Generalized Hough transform

- We want to find a template defined by its reference point (center) and several distinct types of landmark points in stable spatial configuration



Source: S. Lazebnik

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广义霍夫变换

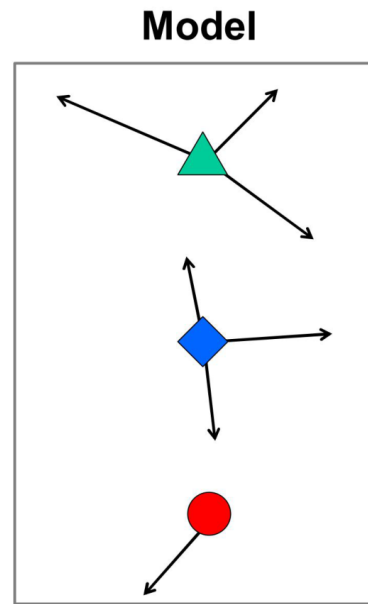
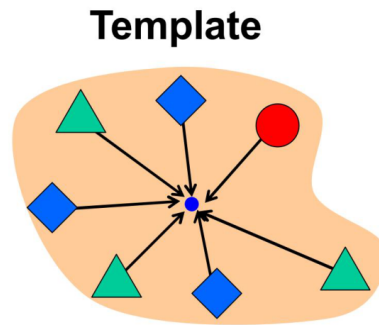
- 我们希望找到一个由其参考点（中心）和若干不同类型的标志点定义的模板，这些标志点处于稳定的空间配置中

模板示意图

来源: S. Lazebnik

Generalized Hough transform

- Template representation: for each type of landmark point, store all possible displacement vectors towards the center



Source: S. Lazebnik

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广义霍夫变换

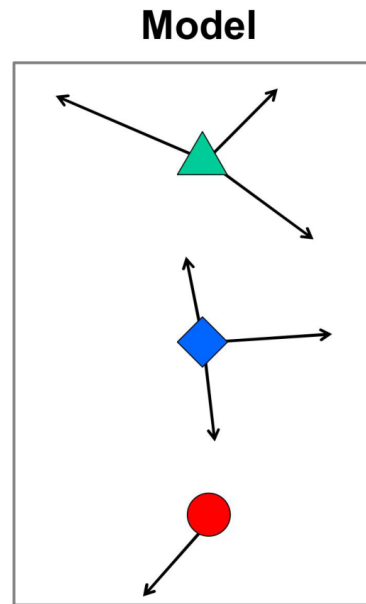
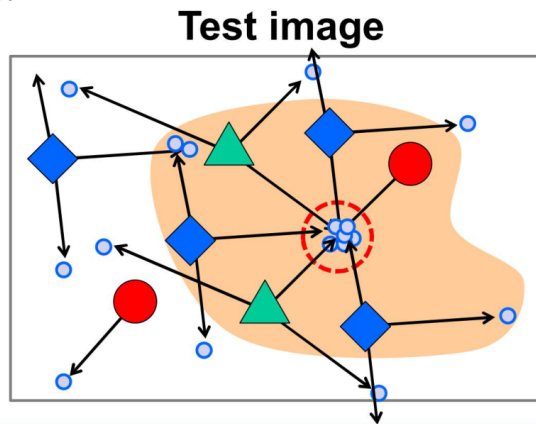
- 模板表示：对于每种类型的标志点，存储其指向中心的所有可能的位移向量

模板示意图 模型

来源：S. Lazebnik

Generalized Hough transform

- Detecting the template:
 - For each feature in a new image, look up that feature type in the model and vote for the possible center locations associated with that type in the model



Source: S. Lazebnik

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广义霍夫变换

- 检测模板:

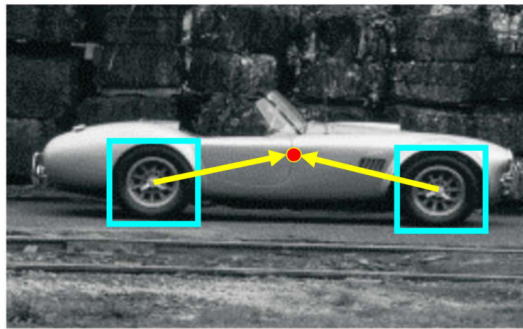
对于新图像中的每个特征，在模型中查找该特征类型，并投票给该类型关联的可能中心位置

测试图像示意图 模型

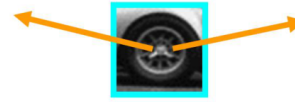
来源: S. Lazebnik

Application in recognition

- Index displacements by “visual codeword”



training image



visual codeword with displacement vectors

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004

Source: S. Lazebnik

2020/3/7

Beijing University of Posts and Telecommunications

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识别中的应用

- 通过“视觉码字”索引位移向量
- 训练图像：提取特征点并关联视觉码字及其位移向量

B. Leibe, A. Leonardis, and B. Schiele, Combined Object Categorization and Segmentation with an Implicit Shape Model, ECCV Workshop on Statistical Learning in Computer Vision 2004

来源：S. Lazebnik

Application in recognition

- Index displacements by “visual codeword”



test image

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004

Source: S. Lazebnik

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识别中的应用

- 通过“视觉码字”索引位移向量
- 测试图像：使用视觉码字匹配并投票确定目标位置

B. Leibe, A. Leonardis, and B. Schiele, Combined Object Categorization and Segmentation with an Implicit Shape Model, ECCV Workshop on Statistical Learning in Computer Vision 2004

来源：S. Lazebnik

Hough transform: Discussion

- Pros
 - Can deal with non-locality and occlusion
 - Can detect multiple instances of a model
 - Some robustness to noise: noise points unlikely to contribute consistently to any single bin
- Cons
 - Complexity of search time increases exponentially with the number of model parameters
 - Non-target shapes can produce spurious peaks in parameter space
 - It's hard to pick a good grid size
- Hough transform vs. RANSAC

Source: S. Lazebnik

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霍夫变换：讨论

优点

- 能够处理非局部性和遮挡
- 可以检测模型的多个实例
- 对噪声具有一定鲁棒性：噪声点不太可能始终对任何一个区间产生一致的投票

缺点

- 搜索时间的复杂度随模型参数数量的增加呈指数增长
- 非目标形状可能在参数空间中产生虚假峰值
- 难以选择合适的网格大小
- 霍夫变换 vs. RANSAC（随机采样一致性）

来源：S. Lazebnik
